31-channel RF Remote Control

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Can you imagine, how convenient your life would be if you could control various appliances inside or outside your house wirelessly. Here is a good news, RF remote controls provide a long range of up to 100 metres and work even through walls. You can control large AC and DC loads and many appliances through relays, including lights, fans, AC systems, home appliances, garage doors, security systems, motor-driven curtains, motorised window blinds, door locks, sprinklers and motorised projection screens.

Here we present a 31-channel RF remote control based on AT89S52 microcontroller. It can switch 'on' and 'off' 31 different devices independently. Any of the 31 outputs can be configured to work independently. There is no need of any encoder and decoder for this 31-channel RF remote control.

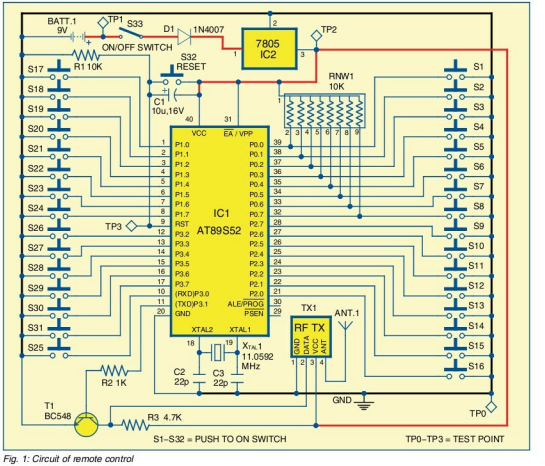


Fig. 1: Circuit of remote control

Circuit and working

The 31-channel remote control consists of three sections, namely, remote control, receiver and appliance driver.

Remote control. Fig. 1 shows the circuit of the remote control. It comprises microcontroller AT89S52 (IC1), ASK transmitter module (TX1), transistor BC548 (T1) and a few discrete components.

The AT89S52 is a low-power, high-performance, 8-bit CMOS microcontroller. It features 8 kB of Flash, 256 bytes of RAM, 32 input/output (I/O) lines, watchdog timer, two data pointers, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry. In addition, the AT89S52 is designed with static logic for operation down to zero frequency and supports two software-selectable power-saving modes. Idle mode stops the CPU while allowing the RAM, timers/counters, serial port and interrupt system to continue functioning. Power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next interrupt or hardware reset.

Test Points

Test point	Details
TP0	Ground (GND) in Fig. 1
TP1	Battery voltage in Fig. 1
TP2	5V in Fig. 1
TP3	High when switch S32 in Fig. 1 pressed
TP4	Ground (GND) in Fig. 2
TP5	5V in Fig. 2
TP6	High when switch S34 in Fig. 2 pressed
TP7	Low when relay RL1 in Fig. 3 energised
TP8	Low when relay RL2 in Fig. 3 energised